T105 - ENGINEERING THERMODYNAMICS
UNIT 1
PART A

1. **Define thermal equilibrium. (NOV 2017)**

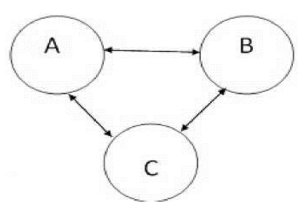
When a system existing in mechanical and chemical equilibrium is separated from its surroundings by a diathermic wall (diathermic means which allows heat to flow) and if there is no spontaneous change in any property of the system the system is said to exist in a state **of thermal equilibrium.**

2. **How the system is classified? (MAY 2019)**

   a. Open system –There is mass transfer across to boundary of system, there may be energy transfer also.
   b. Closed system-There is no mass transfer crosses the system boundary. There may be energy transfer into or out of the system.
   c. Isolated system- There is no mass or energy transfer the across the system boundary.

3. **State the zeroth law of thermodynamics. (MAY 2019, MAY 2016, MAY 2017 NOV 2018, MAY 2015, SEP 2020)**

When a body A is in thermal equilibrium with a body B, and also separately with a body C, then B and C will be in thermal equilibrium with each other.



This is known as the zeroth law of thermodynamics.

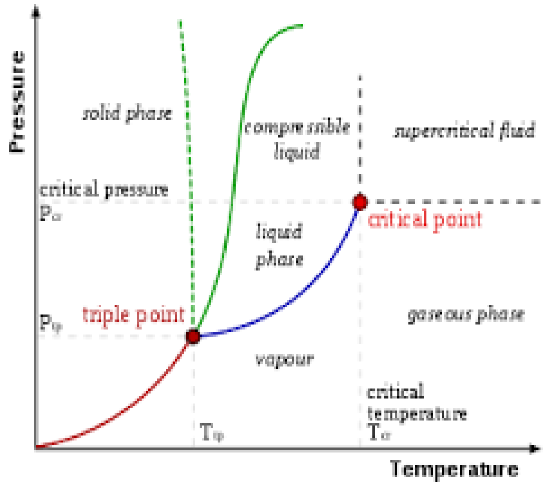
4. **Define thermodynamic equilibrium. (NOV 2016)(NOV 2018 5 mark out of 11 mark)**

A system is said to exist in a state of **thermodynamic equilibrium**, when no change in any macroscopic property is registered, if the system is isolated from its surroundings.

When a system is in thermodynamic equilibrium, it should satisfy the following three conditions.

- Mechanical Equilibrium: Pressure remains constant
- Thermal equilibrium: Temperature remains constant
- Chemical equilibrium: There is no chemical reaction.

5. Draw p-T diagram of pure substance and label all the phases and phase changes.(SEP 2020)



6. Differentiate closed and open system. (NOV 2016, MAY 2017)

S.No.	Closed System	Open System
1	The closed system is a system of fixed mass. There is no mass transfer across the system boundary , there may be energy transfer into or out of the system	The open system is one in which matter crosses the boundary of the system. There may be energy transfer also.
3	Ex: A certain quantity of fluid in a cylinder bounded by a piston	Ex. Air compressor, boiler

7. Define path and process in thermodynamics.(MAY 2016, JAN 2016)

Path

The succession of states passed through during a change of state is called path

Process

When the path is completely specified, the change of state is called a process`

Example: constant volume process

8. Define Quasi static process.(DEC 2015)

The process is said to be quasi – static, it should proceed infinitesimally slow and follows continuous series of equilibrium states. Therefore, the quasi static process may be a reversible process.

9. What is the difference between the classical and the statistical approaches in thermodynamics (DEC 2015, JAN 2016)

S.No.	Statistical approaches:	Classical approaches
1	It deals with the individual	It deals with the overall molecule
2	It requires less number of variables	It requires more number of variables
3	It is also called as microscopic approaches	It is also called as macroscopic approaches
4	It is difficult to calculate	It is easy to calculate
5	Large size molecule is considered	Overall molecule size considered

10. What is the scope of classical thermodynamics? (NOV 2018)

Classical thermodynamics is the description of the states of thermodynamic systems at near-equilibrium that uses macroscopic, measurable properties. It is used to model exchanges of energy, work and heat based on the laws of thermodynamics.

11. What is meant by Continuum? (NOV 2017)

Matter is made up of atoms that are widely spaced in the gas phase. It is convenient to consider the atomic nature of the substance and view it as a continuous homogeneous matter with no holes is called continuum

12. Prove that C_P-C_V=R (MAY 2015)

By 1st Law of thermodynamic

$Q = W + \Delta U$ —— (1)

$Q = mc_p(T_2 - T_1)$ —— (2)

$W = Pdv \rightarrow P(V_2 - V_1)$ $\therefore PV = mRT$

$= mR(T_2 - T_1)$ —— (3)

$\Delta U = mc_v(T_2 - T_1)$ —— (4)

Sub Equ. No. 2, 3 & 4 in (1) Equ.

$mc_p(T_2 - T_1) = mR(T_2 - T_1) + mc_v(T_2 - T_1)$

$= \cancel{mR}$

Dividing throughout by $m(T_2 - T_1)$

$c_p = R + c_v$

$R = c_p - c_v$

Hence proved.

R = Real gas constant

C_p = Specific heat at constant pressure

C_v = Specific heat at constant Volume.

13. What do you mean by intensive and extensive properties?(MAY 2018)

The properties which are independent on the mass of the system is called intensive properties.

e.g., Pressure, Temperature, density etc.,

The properties which are dependent on the mass of the system is called extensive properties. e.g, Volume, Length, Weight etc.

14. Define dryness fraction of liquid vapour mixture.(MAY 2018) (Out of syllabus)

ADDITIONAL QUESTIONS

15. Define thermodynamics properties. (MAY 2018)

Every system has certain characteristics by which its physical condition may be described. E.g. pressure, temperature volume etc. such characteristics are called properties of the system`

16. Define state and thermodynamic cycle? (MAY 2018)

state

When all the properties of a system have definite values, the system is said to exist at a definite state.

Thermodynamic cycle

A thermodynamics cycle is defines as a series of state changes such that the final state is identical with the initial state.

17. What is heat?

Heat is defines as form of energy that is transferred across a boundary by virtue of temperature difference. The temperature difference is the potential or force and heat transfer is the flux.

$$Q = m \times C \times \Delta T \qquad \text{in kJ}$$

1. The following data refer to a 12 cylinder, single acting two-stroke marine diesel engine.
Speed = 150 rpm; Cylinder diameter = 0.8 m; Stroke of piston = 1.2 m;
Area of indicator diagram. = 5.5×10^{-4} m^2, Length of the indicator diagram = 0.06 m and Spring value = 147 MPa/m. Find the net rate of work transfer from the gas to piston in KW. (NOV 2017)

Given data

speed $N = 150$ rpm

$d = 0.8$ m

Stroke $L = 1.2$ m

$a_d = 5.5 \times 10^{-4}$ m^2

$l_d = 0.06$ m

Spring constant = 147 MPa/m

To find

Work transfer W

Sol

Work done $W = P_m L A N$

P_m = mean effective pressure

$$P_m = \frac{a_d}{l_d} \times \text{spring constant}$$

$$= \frac{5.5 \times 10^{-4}}{0.06} \times 147$$

$$P_m = 1.35 \text{ MPa} = 1.35 \times 10^3 \text{ kN/m}^2$$

$$W.D = 1.35 \times 1.2 \times \frac{\pi}{4} d^2 N$$

$$= 1.35 \times 1.2 \times \pi/4 \, (0.8)^2 \times 150$$

$$\boxed{W.D = 122.145 \text{ MJ}}$$

Workdone in one minute = 122.145 MJ/min

for 12 cylinder

$$W = 122.145 \times 12$$

$$= 1465.74 \text{ MJ/min}$$

$$W = 24.429 \text{ MJ/s}$$

$$\boxed{W = 24429 \text{ kW}}$$

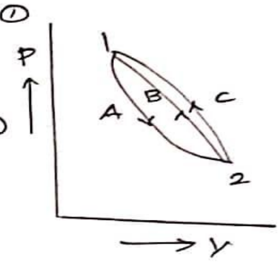
2. Show the internal energy is a property. (MAY 2019)

Consider a system which changes its state from state 1 to state 2 by following the path A, and returns from state 2 to state 1 by following the path B so the system undergoes a cycle. writting first law for path A

$$Q_A = \Delta E_A + W_A \quad \text{—(1)}$$

and for path B

$$Q_B = \Delta E_B + W_B \quad \text{—(2)}$$



the process A and B together constitute a cycle for which

$$(\Sigma W)_{cycle} = (\Sigma Q)_{cycle}$$

$$W_A + W_B = Q_A + Q_B$$

$$Q_A - W_A = W_B - Q_B \quad \text{—(3)}$$

from Equ. No 1, 2, and 3

$$\Delta E_A = -\Delta E_B \quad \text{—(4)}$$

Similarly had the system returned from state 2 to state 1 by following the path C instead of path B

$$\Delta E_A = -\Delta E_C \quad \text{—(5)}$$

from Equ No. 4 & 5

$$\Delta E_B = \Delta E_C$$

From above Equation it can be concluded that energy change in path B and path C are equal and hence on the end states.

It has been already shown that all the properties are point functions and hence energy is also a property of the system.

3. Explain how the Zeroth law of thermodynamics can be used for temperature measurement. (MAY 2019, NOV 2017)

The property which distinguishes thermodynamics from other sciences is temperature. One might say that temperature bears as important a relation to thermodynamics as force does to statics or velocity does to dynamics.

Temperature is associated with the ability to distinguish hot from cold. When two bodies at different temperatures are brought into contact, after some time they attain a common temperature and are then said to exist in thermal equilibrium.

"When a body A is in thermal equilibrium with a body B, and also separately with a body C, then B and C will be in thermal equilibrium with each other".

This is known as the zeroth law of thermodynamics. It is the basis of temperature measurement.

In order to obtain a quantitative measure of temperature, a reference body is used, and a certain physical characteristic of this body which changes with temperature is selected. The changes in the selected characteristic may be taken as an indication of change in temperature.

The selected characteristic is called the thermometric property, and the reference body which is used in the determination of temperature is called the thermometer. A very common thermometer consists of a small amount of mercury in an evacuated capillary tube, in this case the extension of the mercury in the tube is used as the thermometric property.

4. A fluid is confined in a piston and cylinder machine, passes through a complete cycle of four processes. The sum of all heat transferred during a cycle is - 340 kJ. The system completes 200 cycles per min. Complete the following table showing the methods for each item and compute the net rate of work kW. (NOV 2016)

Process	Q (kJ/min)	W (kJ/min)	AE (kJ/min)
1 — 28	0	4340	-
2 — 3	42000	0	-
3 — 4	-4200	-	-73200
4 — 1	-	-	-

given data

$dQ = -340 \text{ kJ}$

system complete 200 cycle per mins.

To find

dW, complete the table.

sol

~~Process~~

$dQ = -340 \text{ kJ}$

200 cycle per min.

$dQ = -340 \times 200$

$dQ = -68000 \text{ kJ/min}$

Proces 1-2

$Q_{1-2} = W_{1-2} + \Delta E_{1-2}$

$0 = 4340 + \Delta E$

$\boxed{\Delta E_{1-2} = -4340 \text{ kJ/min}}$

Process 2-3

$Q_{2-3} = W_{2-3} + \Delta E_{2-3}$

$42000 = 0 + \Delta E$

$\boxed{\Delta E_{2-3} = 42000 \text{ kJ/min}}$

Process 3-4

$Q_{3-4} = W_{3-4} + \Delta E_{3-4}$

$-4200 = W - 73200$

$\boxed{W_{3-4} = 69000 \text{ kJ/min}}$

$$dQ = Q_{1-2} + Q_{2-3} + Q_{3-4} + Q_{4-1}$$

$$-68000 = 0 + 42000 - 4200 + Q_{4-1}$$

$$\boxed{Q_{4-1} = -105800 \text{ kJ/min}}$$

First Law of thermodynamics for cycling process

$$dQ = dW$$

$$dW = -68000 \text{ kJ/min.}$$

$$dW = W_{1-2} + W_{2-3} + W_{3-4} + W_{4-1}$$

$$-68000 = 4340 + 0 + 69000 + W_{4-1}$$

$$\boxed{W_{4-1} = -141340 \text{ kJ/min}}$$

by 1st Law of Thermodynamics.

$$Q_{4-1} = W_{4-1} + \Delta E_{4-1}$$

$$\boxed{\Delta E_{4-1} = 35540 \text{ kJ/min}}$$

Since $dQ = dW$

$$dW = -68000 \text{ kJ/min}$$

$$= \frac{-68000}{60}$$

$$\boxed{dW = -1133.3 \text{ kW}}$$

5. A fluid system undergoes a non-flow frictionless process following the pressure-volume relation as $P=\frac{5}{V}+1.5$ where p is in bar and V is in m^3. During the process the volume changes from 0.15 m^3 to 0.05 m^3 and the system rejects 45 kJ of heat. Determine: (a) Change in internal energy; (b) Change in enthalpy.(NOV 2016)

Given data

$P = 5/V + 1.5$ in bar

$= (5/V + 1.5) \times 10^2$ kN/m^2

$V_1 = 0.15 m^3$

$V_2 = 0.05 m^3$

$Q = -45$ kJ

To find

ΔU, ΔH = ?

Sol

$W = ?$

$$W = \int_{V_1}^{V_2} P \, dV$$

$$= \int_{V_1}^{V_2} \frac{500}{V} + 150 \, dV$$

$$= 500 \int_{V_1}^{V_2} \frac{1}{V} dV + 150 \int_{V_1}^{V_2} dV$$

$$= 500 \left[\ln(V)_{V_1}^{V_2} \right] + 150 [V]_{V_1}^{V_2}$$

$$= 500 \ln(V_2 - V_1) + 150 (V_2 - V_1)$$

$$= 500 \ln(V_2/V_1) + 150 (V_2 - V_1)$$

$$= 500 \ln(0.05/0.15) + 150 (0.05 - 0.15)$$

$$\boxed{W = -564 \text{ kJ}}$$

Change in Internal Energy

$$Q = W + \Delta U$$

$$-45 = -564 + \Delta U$$

$$\boxed{\Delta U = 519\ KJ}$$

Change in Enthalpy

$$\Delta H = h_2 - h_1$$

$$h_1 = U_1 + P_1 V_1$$

$$h_2 = U_2 + P_2 V_2$$

$$\Delta H = U_2 + P_2 V_2 - U_1 + P_1 V_1$$

$$= U_2 - U_1 + (P_2 V_2 - P_1 V_1)$$

$$\Delta H = \Delta U + P_2 V_2 - P_1 V_1$$

$$P_1 = ? \qquad P_2 = ?$$

$$P = 500/V + 150$$

$$P_1 = 500/V_1 + 150$$

$$P_2 = 500/V_2 + 150$$

$$P_1 = 500/0.15 + 150 = 3483\ KN/m^2$$

$$P_2 = 500/0.05 + 150 = 10150\ KN/m^2$$

$$\Delta H = \Delta U + P_2 V_2 - P_1 V_1$$

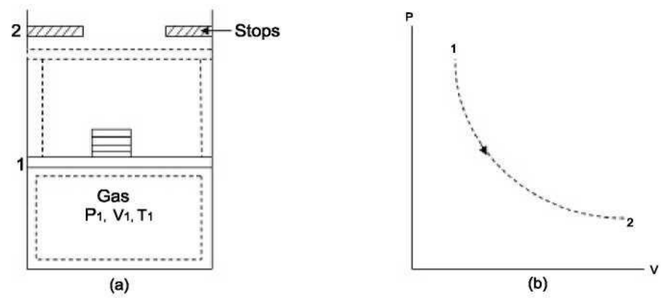
$$= 519 + (10150 \times 0.05) - (3483 \times 0.15)$$

$$\boxed{\Delta H = 504.05\ KJ}$$

6. Discuss quasi-static process with neat diagram and plot the quasi-static process in a P-V plot. (MAY 2016)

A quasi static process is defined as a process in which the properties of the system depart extremely small from the thermodynamic equilibrium path.
If the properties of the system has finite departures from thermodynamic equilibrium path the process is said to be non-quasi static.

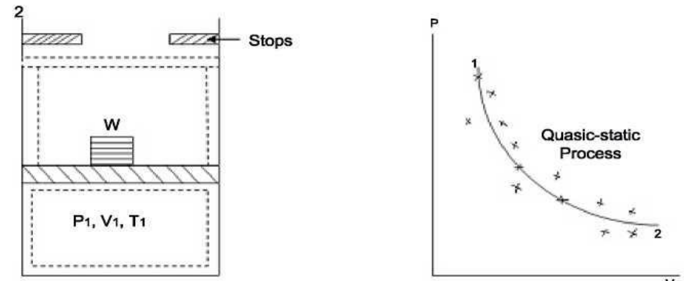
To understand the above concept let us consider a frictionless piston-cylinder arrangement having gas at state 1 defined by (P_1,V_1,T_1). A system is in thermodynamic equilibrium due to external weight W and it is isolated.



Non quasi static process

If a weight is removed there will be large unbalanced pressure forces between the system and surroundings. Due to this the gas will suddenly expand and the piston will move up till it hits the stop provided it the cylinder. The system will reach to new equilibrium state defined by (p_2, V_2, T_2) but the intermediate states of the system cannot be defined since it has passed through non-equilibrium states. The process is said to be non-quasi static and it is represented by dotted curve. Now consider the above case in which the weight W, is made up of extremely small amount of weights. If the weight is removed in succession gradually from the top of piston the gas will expand slowly and it will pass through large number of thermodynamic state till the system again reaches to state 2.

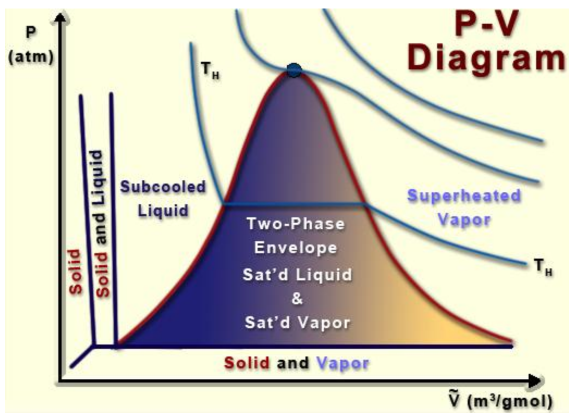
In this case the departure of each state of the system from thermodynamic equilibrium state is extremely small and the locus of such thermodynamic equilibrium states represented by process path by solid lines. Such a process is said to be a quasi static.



Quasistatic process

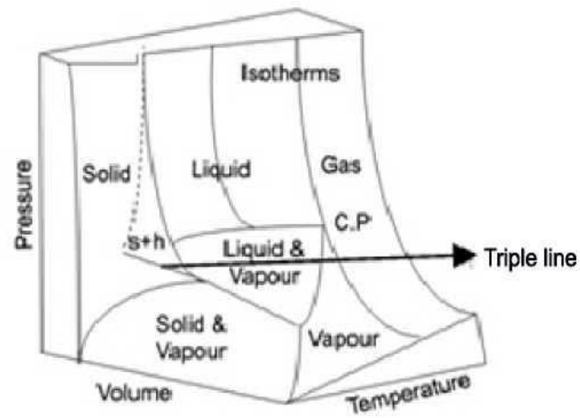
7. Detail the P-V diagram for a pure substance. Plot P-V-T diagram of water and pure substance other than water.(MAY 2016, SEP 2020)

- In the p-V diagram the saturated liquid curve and the saturated vapour curve meets at the critical point the regions left to the saturated liquid line is compressed liquid region the region right to the saturated vapour line is called super heated vapour region.
- In between two line is the to take liquid and vapour region there are several isotherms in the p V diagram, the critical isotherm has a zero slope at critical point .
- The isotherms below the critical point experiences discontinued in slope as they cross the saturated liquid and saturated vapour line.
- In between these two lines the slope of isotherm is zero.



p V diagram

A pure substance is one which has a homogeneous and invariable chemical composition even though there is a change of phase. For example liquid water, mixture of water and steam, mixture of ice and water. However mixture of air and water is not a pure substance since it has variable chemical composition.



P-V-T Surface for pure substance (water) the heating process reveals that the temperature of solids raises and then during the phase of change fro solid to liquid or solid to vapor the temperature remains constant. This phenomenon is common to all phase changes. Since the temperature is constant pressure and temperature are not independent properties and cannot be used to specify state during a phase of change.

This figure illustrates the equilibrium states for a pure substance which expands on fusion. All the equilibrium states lie on the surface of the model. States represented by the space above or below the surface are not possible. It can be seen that the triple point appears as a line in this representation. The critical point has no liquid phase exists at temperatures above the isotherms through this point. At the critical point the temperatures and pressures are called critical temperatures and critical pressures. When the temperatures of the substance is above the critical value it is called a gas. It is not possible to cause a phase change in a gas unless the temperature is lower to a value less than the critical temperature.

8. Explain different types of equilibrium, efficiencies, systems, heat transfer(SEP 2020)

Types of equilibrium

A system is said to exist in a state of **thermodynamic equilibrium**, when no change in any macroscopic property is registered, if the system is isolated from its surroundings.

When a system is in thermodynamic equilibrium, it should satisfy the following three conditions.

- Mechanical Equilibrium: Pressure remains constant

  When there is no change in force or pressure with time is called mechanical equilibrium.
- Thermal equilibrium: Temperature remains constant

  There is no change in temperature with time is called thermal equilibrium
- Chemical equilibrium: There is no chemical reaction.

  There is no chemical reaction are transferred of matter from one part to another part such as diffusion, solution.

Energy Efficiencies

Energy efficiencies is using less energy efficiency to provide same outout

Energy

Energy is the ability to bring about change or to do work. Thermodynamics is the study of energy. First law of thermodynamics: energy can be changed from one form to another, but it cannot be created or destroyed.

Systems

System can be defines as quantity of matter or region that take into be consider for analyses

Open system

When a system has both mass and energy transfer it is called as open system.

Example: Air Compressor.

Closed system

When a system has only heat and work transfer, but there is no mass transfer, it is called as closed system.

Example: Piston and cylinder arrangement.

Isolated system

Isolated system is not affected by surroundings. There is no heat; work and mass transfer take place. In this system total energy remains constant.

Example: Entire Universe

Heat transfer

Heat is defines as form of energy that is transferred across a boundary by virtue of temperature difference. The temperature difference is the potential or force and heat transfer is the flux.

$$Q = m \times C \times \Delta T \qquad \text{in kJ}$$

9. A gas initially at 10 bar, 0.01 m^3 expand according to straight line variation upto the filial state of 1 bar, 0.05 m^3. The law of process is given by p=aV+b, where 'a' and 'b' are constants.-Determine the work done during the Quasi static process.(JAN 2015)

given data

$P_1 = 10 \text{ bar} = 1000 \text{ kN/m}^2$

$V_1 = 0.01 \text{ m}^3$

$P_2 = 1 \text{ bar} = 100 \text{ kN/m}^2$

$V_2 = 0.05 \text{ m}^3$

To find

W.D = ?

Sol

$P = a + bv$

$P_1 = a + bv_1$ ——— (1)

$P_2 = a + bv_2$ ——— (2)

$1000 = a + b\,0.01$

$-100 = a + b\,0.05$

$900 = -0.04b$

$b = -900/0.04$

$b = -22500$

sub b in equ. (1)

$1000 = a + b\,0.01$

$1000 = a + (-22500 \times 0.01)$

$a = 1225$

$$W = \int_{v_1}^{v_2} P\,dv$$

$$= \int_{v_1}^{v_2} (a + bv)\,dv$$

$$= \int_{v_1}^{v_2} 1225 + (-22500)\,v\,dv.$$

$$= 1225 \int_{v_1}^{v_2} dv + -22500 \int_{v_1}^{v_2} v\,dv$$

$$= 1225 \left[V \right]_{V_1}^{V_2} - 22500 \left[V^2/2 \right]_{V_1}^{V_2}$$

$$= 1225 \left[V_2 - V_1 \right] - 22500 \left[\frac{V_2^2 - V_1^2}{2} \right]$$

$$= 1225 \left(0.05 - 0.01 \right] - 22500 \left[\frac{0.05^2 - 0.01^2}{2} \right]$$

$$\boxed{W.D = 22 \text{ KJ}}$$

10. In a piston cylinder arrangement the pressure inversely proportional to the square of volume. The initial pressure is 10 bar in the cylinder and initial volume is 0.1 m^3. The volume is now changed so that the final pressure is 2 bar. Find the work done in KJ.(JAN 2015)

given data

The Relation = $P \propto \frac{1}{V^2}$

$P_1 = 10\,bar = 1000\ KN/m^2$

$V_1 = 0.1\,m^3$

$P_2 = 2\,bar = 200\ KN/m^2$

To find

W.D

Sol

Given relation

$P \propto \frac{1}{V^2}$

$P = K \cdot \frac{1}{V^2}$

$P_1 = K \cdot \frac{1}{V_1^2}$

$P_2 = K \cdot \frac{1}{V_2^2}$

$10 = K \cdot \frac{1}{(0.1)^2}$

$K = 10$

$P_2 = K \cdot \frac{1}{V_2^2}$

$200 = 10 \frac{1}{V_2^2}$

$V_2^2 = \sqrt{10/200}$

$V_2 = 0.223\,m^3$

$$W = \int_{V_1}^{V_2} P\,dV$$

$$= \int_{V_1}^{V_2} \frac{K}{V^2}\,dV$$

$$W.D = K\left[\frac{1}{V_1} - \frac{1}{V_2}\right]$$

$$= 10\left[\frac{1}{0.1} - \frac{1}{0.233}\right]$$

$$\boxed{W.D = 55.1\,KJ}$$

11. A fluid is confined in a cylinder by a spring-loaded, frictionless piston so that the pressure in the fluid, is a linear function of the volume (p = a + bV). The internal energy of the fluid is given by U = (34 + 3.15 pv) where U is in kJ, p in kPa and V in cubic meter. If the fluid changes from an initial state of 170 kPa, 0.03 m^3 to a final state of 400 kPa. 0.06 m^3, with no work other than that done on the piston, find the direction and magnitude of the work and heat transfer.(JAN 2016)

Given data,

$P = a + bV$

$U = 34 + 3.15\ PV$

$P_1 = 170\ kPa \rightarrow 170\ kN/m^2$

$P_2 = 400\ kPa \rightarrow 400\ kN/m^2$

$V_1 = 0.03\ m^3$

$V_2 = 0.06\ m^3$

To find,

W and Q (direction of magnitude)

$W = \int_{V_1}^{V_2} p\,dv$

$P = a + bV$

$P_1 = a + bV_1 \rightarrow (1)$

$P_2 = a + bV_2 \rightarrow (2)$

$170 = a + b(0.03)$

$\underline{\overset{(-)}{}400 = \overset{(-)}{}a \overset{(-)}{+} b(0.06)}$

$-230 = -b(0.03)$

$\boxed{b = 7666.6} \rightarrow (3)$

Sub (3) in (1)

$170 = a + 7666.6(0.03)$

$\boxed{a = -59.9}$

$W = \int_{V_1}^{V_2} P\,dv$

$$= \int_{V_1}^{V_2} (-59.9 + 7666.6v)\,dv$$

$$= \int_{V_1}^{V_2} (-59.9\,dv) + \int_{V_1}^{V_2} (7666.6v)\,dv$$

$$= -59.9(V_2 - V_1) + 7666.6\left(\frac{V_2^2 - V_1^2}{2}\right)$$

$$= -59.9(0.03) + 3833.3(3.6 - 0.9)10^{-3}$$

$$\boxed{W = 8.6\ kJ}$$

$$U = 34 + 3.15\,PV$$

$$U_1 = 34 + 3.15\,P_1V_1$$

$$= 34 + 3.15(170)(0.03)$$

$$U_1 = 50.06\ kJ$$

$$U_2 = 34 + 3.15\,P_2V_2$$

$$= 34 + 3.15(400)(0.06)$$

$$U_2 = 109.6\ kJ$$

$$\Delta U = U_2 - U_1$$

$$= 109.6 - 50.06$$

$$\boxed{\Delta U = 59.53\ kJ}$$

$$Q = W + \Delta U$$

$$= 8.6 + 59.53$$

$\boxed{Q = 68.08\ kJ}$ Q Absorb by the System

12. Define the following terms : (a) Thermodynamics (b) Microscopic Approach, (c) Continuum.(JAN 2016)

Thermodynamics

Thermodynamics is the science of energy transfer and its effect on the physical properties of substances. It is based upon observations of common experience which h ive been formulated into thermodynamic laws. These laws govern the principles of energy conversion. The applications ofthe thermodynamic laws and principles are found in all fields of energy technology, notably in steam and nuclear powerplants, interna! combustion engines, gas turbines, air conditioning, refrigeration, gas dynamics, jet propulsion, compressors, chemical process plants, and direct energy conversion devices.

Microscopic Approaches

The microscopic point of view, matter is composed of myriads of molecules. If it is a gas, each molecule at a given instant has a certain position, velocity, and energy, and for each molecule these change very frequently as a result of collisions.

Continuum

Matter is made up of atoms that are widely spaced in the gas phase. It is convenient to consider the atomic nature of the substance and view it as a continuous homogeneous matter with no holes is called continuum

13. An engine cylinder has a piston of area 0.1.2 m^3 and contains gas at a pressure of 1.5 MPa. The gas expands according to a process which is represented by a straight line on a pressure - volume diagram. The final pressure is 0.15 MPa. Calculate the work done by a gas on the piston if the stroke is 0.30 m.(NOV 2018)

Given data

$P_1 = 1.5 \text{ MPa} = 1.5 \times 10^3 \text{ kN/m}^2$

$P_2 = 0.15 \text{ MPa} = 0.15 \times 10^3 \text{ kN/m}^2$

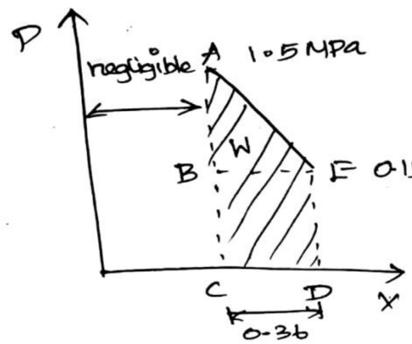
Area (A) = 0.12 m^2

Length (L) = 0.3 m

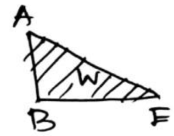
To find

$W = ?$

Sol

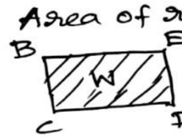


Area of triangle



AB x BE

Area of rectangle



BE x CD

Volume of cylinder

= area x stroke.

$= 0.12 \times 0.3$

$V = 0.036 \text{ m}^3$

Workdone = area of traiangle + area of reetangle.

$= \frac{1}{2}(b \times h) + (b \times h)$

$= \frac{1}{2}[BE \times AB] + [CD \times BE]$

AB = length different in pressure $P_2 - P_1$

$= (1.5 - 0.15) \times 10^3$

BE = height of final pressure P_2.

$= 0.15 \times 10^3$

$W = \frac{1}{2}[0.036 \times [(1.5 - 0.15) \times 10^3] + [0.036 \times 0.15 \times 10^3]$

$\boxed{W = 29.7 \text{ KJ}}$

14. A pump discharge a liquid into drum at the rate of .032 m^3/s. The drum has a diameter of 1.5 m, a length of 4.2 m and can hold 3000 kg of the liquid. Find the density of the liquid, the mass flow rate and the time required to fill the tank. (MAY 2017)

Given data

mass $M = 3000\ kg$

diameter $d = 1.50 m$

$L = h = 4.20\ m$

Volume flow rate $V_f = 0.32\ m^3/s$

To find

m, ρ

Sol

Volume of drum $= \pi r^2 h$

$= \pi \times \left(\frac{1.50}{2}\right)^2 \times 4.20$

$V = 7.427 m^3$

density $\rho = \frac{mass}{volume}$

$= \frac{3000}{7.427}$

$\boxed{\rho = 404.203\ kg/m^3}$

mass flow rate

= Volume flow rate * density

$= 0.032 \times 404.203$

$\boxed{m_f = 12.9345\ kg/s}$

15. **Water at 90°C flowing at the rate of 2 kg/s mixes adiabatically with another stream of water at, 30°C flowing at the rate of 1 kg/s. Estimate the entropy generation rate and the rate of energy loss due to mixing. Take To = 300 K. (NOV 018) (Out of syllabus)**

16. **Determine the quality or temperature of the following substances in the given states :**
    a) **Ammonia, 26°C and 0.074 m^3/kg**
    b) **Ammonia, 550 KPa and 0.31 m^3/kg**
    c) **Freon 12,0.35 MPa and 0.036 m^3/kg**
    d) **Methane,0.5 MPa and 1.0 m^3/K mol (MAY 2017) (Out of syllabus)**

17. **An air compressor requires a shaft work of 200 kJ/kg of air and the compression of air causes increases the enthalpy of air by 100 kJ/kg of air. Cooling water required for cooling the compressor picks up heat by 90 kJ/kg of air. Determine the heat transferred from compressor to atmosphere. (MAY 2019) (Refer second unit, problem no.15)**

18. **Steam initially at 0.3 MPa, 250^{o}C is cooled at constant volume. Determine (a) temperature at which the steam becomes saturated vapour, (b) steam quality at 80^{o}c and (c) heat transferred per kg of steam while cooling from 250^o C to 80^o C. (MAY 2018) (out of syllabus)**

ADDITIONAL PROBLEM

1. **A stationary mass of gas is compressed without friction from an initial state of 0.3 m^3 and 0.105MPa to a final state of 0.15 m^3 and 0.105 MPa, the pressure remaining constant during the process. There is a transfer of 37.6 k-J of heat from the gas during the process. How' much does the internal energy of the gas change?**

Given data

$V_1 = 0.3 m^3$

$P_1 = 0.105\ mpa = 105\ kn/m^2$

$V_2 = 0.15 m^3$

$P_2 = 0.105 mpa = 105\ kn/m^2$

$Q = -37.6\ kJ$

To find

$\Delta U = ?$

Sol

$Q = W + \Delta U$

$W = \int_1^2 P dv$

$= P(V_2 - V_1)$

$= 0.105 \times 10^3 (0.15 - 0.3)$

$\boxed{W = -15.75\ kJ}$

$Q = W + \Delta U$

$-37.6 = -15.75 + \Delta U$

$\Delta U = -37.6 + 15.75$

$\boxed{\Delta U = -21.85\ kJ}$

Result:-

$\Delta U = -21.85\ kJ$

2. **A spherical balloon of 30cm diameter compressed air at a pressure of 1.5 bar. The diameter of balloon is increased to 40cm by heating and the process the pressure is proportional to its diameter. Calculate the work done assuming the process to be quasic static.**

Given data

$D_1 = 30cm = 0.3m$

$D_2 = 40cm = 0.4m$

$P \propto D$

$P_1 = 1.5 \text{ bar} = 150 \text{ kn/m}^2$

To find

W.D

Sol

$P \propto D$

$P = K \cdot D$

$P_1 = K \cdot D_1$

$P_2 = K \cdot D_1$

$P_1 = K D_1$

$150 = K \cdot 0.3$

$\underline{\underline{K = 500}}$

$P_2 = K \cdot D_2$

$P_2 = 500 \times 0.4$

$P_2 = 200 \text{ kn/m}^2$

$$W.D = \int_{V_1}^{V_2} P\,dV$$

$$= \int_{V_1}^{V_2} K \cdot D\,dV$$

Volume of balloon

Sphere $V = 4/3 \pi r^3$

$$= 4/3 \pi (D/2)^3$$

$$= \cancel{4}/3 \pi \frac{D^3}{\cancel{8}\,2}$$

$$V = \frac{\pi D^3}{6}$$

$$dV = \frac{\pi\, 3D^2 \cdot dD}{6}$$

$$dV = \frac{\pi D^2}{2} \cdot dD$$

$$W = \int_{D_1}^{D_2} K \cdot D \cdot \frac{\pi D^2}{2} d \cdot D$$

$$= \frac{K\pi}{2} \int_{D_1}^{D_2} D^3 \, dD$$

$$= \frac{K\pi}{2} \left[\frac{D^4}{4} \right]_{D_1}^{D_2}$$

$$= \frac{K\pi}{2} \left[\frac{D_2^{\,4} - D_1^{\,4}}{4} \right]$$

$$= \frac{500 \times \pi}{2} \left[0.4^4 - 0.3^4 \right]$$

$$\boxed{W.D = 3.43 \text{ KJ}}$$

3. Air is compressid in a cylinder from $0.6m^3$ to $0.1m^3$ by a piston. The relation between the pressure and volume is given by p = 5-3V where p is in bar and v is in m^3. Compute the magnitude of workdone on the system.

Given data

$$V_1 = 0.6\ m^3$$
$$V_2 = 0.1\ m^3$$
$$P = (5-3v) \times 100\ kN/m^2$$

To find

$$W = ?$$

Sol

$$W.D = \int_{v_1}^{v_2} P\,dv$$
$$= \int_{v_1}^{v_2} (5-3v)\,dv$$
$$= 5\int_{v_1}^{v_2} dv - 3\int_{v_1}^{v_2} v\,dv$$
$$= 5[v]_{v_1}^{v_2} - 3[v^2/2]_{v_1}^{v_2}$$
$$= 5[v_2 - v_1] - 3\left[\frac{v_2^2 - v_1^2}{2}\right]$$
$$= 5[0.1-0.6] - 3\left[\frac{0.1^2 - 0.6^2}{2}\right]$$
$$W.D = -1.975 \times 100$$
$$\boxed{W.D = -197.5\ kJ}$$